SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING) Department of Computer Science and Engineering

ASSESSMENT TOOL 1 – APPLICATION BASED QUESTIONS

Course Code:	CSA06	**Course Title:**	Design and Analysis of Algorithms
CO Assessed:	CO1 – Precision (BL1, BL2, BL3)	**Assessment:**	Assessment Tool 1 – Application Based Questions
Weightage:	40% of CIA	**Total Marks:**	100 (2 Questions × 50 Marks)
CO1 Statement:	*Determine algorithm efficiency using asymptotic notations and mathematical techniques including Master Theorem and substitution method. (BL1, BL2, BL3)*		
Student Name:	Y. Jithendra Reddy	**Reg. No.:**	192525303
Section / Batch:	C	**Date:**	22.07.2026

Instructions to Students

- Answer BOTH questions. Each question carries 50 Marks. Total: 100 Marks.
- Clearly show all steps in derivations and complexity analysis.
- Use proper asymptotic notation (Big-O, Big-Θ, Big-Ω) wherever required.
- Justify each step in recurrence solving using appropriate methods.
- Neat presentation and logical explanation are mandatory

Application-Based Questions	Apply Master Theorem and asymptotic analysis to real-world scenarios	Q1 – Q2	Q1 –50 Q2 –50

SECTION A – APPLICATION BASED QUESTIONS

Code of Conduct

I Y. Jithendra Reddy (192525303)) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _Y. Jithendra Reddy

RUBRICS FOR CALCULATION

Understandin g of Problem Context	20%	Clearly understands the problem and accurately identifies all requirements	Good understanding with minor gaps	Basic understanding; some requirements unclear	Poor understanding or misinterpretation of the problem
Application of Concepts	30%	Accurately applies appropriate concepts to solve complex problems	Applies relevant concepts correctly with minor errors	Applies basic concepts with limited accuracy	Fails to apply appropriate concepts
Problem- Solvi ng Approach	20%	Logical, well-structured, and efficient approach	Mostly logical approach with minor gaps	Basic approach with some inconsistencies	Illogical or unclear approach
Accuracy of Solution	20%	Completely correct solution with proper justification	Mostly correct with minor errors	Partially correct solution	Incorrect or incomplete solution
Clarity	10%	Clear, well-organized, and properly explained solution	Understandable with minor clarity issues	Limited clarity and explanation	Poorly presented and difficult to understand

Marks Evaluation:

Question No.	Understandin g of Problem Context (20%)	Application of Concepts (30%)	Problem- So lving Approach (20%)	Accuracy of Solution (20%)	Clarity (10%)	Total Marks
Q1 (50 Marks)						
Q2 (50 Marks)						
					Overall Total (100)	

Faculty In-charge	Course Coordinator	Head of Department
Signature: ________________	Signature: ________________	Signature: ________________
Date: ____________________	Date: ____________________	Date: ____________________

S.No	Register No.	Student Name	Assigned Question 1	Assigned Question 2
1	192311215	MUTHU VAISHNAVI T	Q1	Q2
2	192311229	JENIECATHRINA J	Q3	Q4
3	192324152	ABINAYA SRI N	Q5	Q6
4	192324287	PRAVEEN KUMAR R	Q7	Q8
5	192372122	NADENDLA THANVEER BASHA	Q9	Q10
6	192372336	SIGI SREENIVASULU	Q11	Q12
7	192411025	SRINIVASAN A	Q13	Q14
8	192411031	GUDIPALLI VEMULA SHABARISH	Q15	Q16
9	192411039	GOKUL RAJAN R	Q17	Q18
10	192411061	GANGADEVI LOKESH	Q19	Q20
11	192411156	KEERTHIPATI HEMA SREE	Q21	Q22
12	192411249	CHENNABOINA BHAVYA SRI	Q23	Q24
13	192421030	NAVEEN KUMAR S	Q25	Q26
14	192421126	MOHAMMED ANAS M	Q27	Q28
15	192421259	JAYAKRISHNA V	Q29	Q30
16	192421333	PUSHPAVALLI V	Q31	Q32
17	192424016	DINESH M S	Q33	Q34
18	192424174	YOKESH KANNA R	Q35	Q36
19	192424259	PERA HEMIKA	Q37	Q38
20	192424327	CHARULATHA S	Q39	Q40
21	192424352	R SAI TEJA	Q41	Q42
22	192465048	POTHURAJU SAI CHARAN TEJ	Q43	Q44
23	192472087	MANCHIKALAPATI VISHNU VARDHAN	Q45	Q46
24	192472149	CHINNI VENKATA NAGA SAI POORNA TEJA	Q47	Q48

25	192472307	SHAIK AZEEM	Q49	Q50
26	192511060	Manoj M	Q51	Q52
27	192521075	Kaviya R	Q53	Q54
28	192521121	Srimathi S	Q55	Q56
29	192521352	Jagadish V	Q57	Q58
30	192524078	Rafeez Ahamed S I	Q59	Q60
31	192524237	Athin Siva S	Q61	Q62
32	192525202	Muthu Neelakanta Naidu	Q63	Q64
33	192525289	B Chandu	Q65	Q66
34	192525303	Yedururu Jithendra Reddy	Q67	Q68
35	192565016	Divya Sree P	Q69	Q70
36	192565069	Yaswanthika Shree S	Q71	Q72
37	192571005	V Sanjana	Q73	Q74
38	192571047	Shaffria A	Q75	Q76
39	192571051	Podalapalli Darshith	Q77	Q78
40	192572106	Theshon Mishmaan M	Q79	Q80

Q67. Application: Digital Dictionary Search Engine
A dictionary search system reduces the remaining search entries by half after each comparison. Define divide-and-conquer strategy, explain how input size reduces, and calculate the number of steps. Determine both the time complexity and space complexity of this search process.

Problem Statement
A dictionary search system reduces the remaining search entries by half after each comparison. This follows a **divide-and-conquer strategy**.
1. Divide-and-Conquer Strategy
Divide-and-conquer is a technique where a problem is divided into smaller subproblems and solved recursively.
In dictionary search:

- The middle entry is compared with the target word.
- If the word is smaller, the left half is searched.
- If the word is larger, the right half is searched.
- This process continues until the word is found.

2. Input Size Reduction
The search space reduces as:

$$n, \frac{n}{2}, \frac{n}{4}, \frac{n}{8}, \dots$$

After k comparisons:

$$\frac{n}{2^k} = 1$$

Therefore,

$$k = \log_2 n$$

The maximum number of steps required is:

$$\boxed{\log_2 n}$$

3. Time Complexity

The recurrence relation is:

$$T(n) = T(n/2) + 1$$

Comparing with Master Theorem:

$$T(n) = aT(n/b) + f(n)$$

where:

$$a = 1,\ b = 2,\ f(n) = 1$$
$$n^{\log_b a} = n^{\log_2 1} = 1$$

This gives:

$$\boxed{T(n) = \Theta(\log n)}$$

4. Space Complexity

For recursive implementation, the maximum number of recursive calls is:

$$\log_2 n$$

Each call requires constant memory for variables and returns information.
Therefore:

$$\boxed{Space\ Complexity = O(\log n)}$$

Final Answer

Time Complexity:

$$\boxed{\Theta(\log n)}$$

Space Complexity:

$$\boxed{O(\log n)}$$

Conclusion: The dictionary search is efficient because it eliminates half of the entries after every comparison, resulting in logarithmic time complexity.

Q68. Application: Cloud Backup Analysis System
A cloud system processes backup data using the recurrence T(n) = T(n/2) + n. Recall recurrence solving techniques, explain how the function grows, and determine the asymptotic time complexity using an appropriate method. Also analyze the space complexity of the recursive calls.

Problem Statement
A cloud system processes backup data using the recurrence:

$$T(n) = T(n/2) + n$$

The system reduces the data size by half while performing linear processing work at each step.

1. Identify Parameters
Using Master Theorem:

$$T(n) = aT(n/b) + f(n)$$

Compare:

$$T(n) = T(n/2) + n$$

Therefore:

$$a = 1,\ b = 2,\ f(n) = n$$

2. Recursion Tree Analysis
The work performed at each level is:
Level 0:

$$n$$

Level 1:

$$n/2$$

Level 2:

$$n/4$$

Total work:

$$n + \frac{n}{2} + \frac{n}{4} + \ldots$$

This is a geometric series whose total value is:

$$2n$$

Hence:

$$T(n) = \Theta(n)$$

3. Master Theorem Application
Calculate:

$$n^{\log_b a} = n^{\log_2 1} = 1$$

Since:

$$f(n) = n$$

is larger than:

$$n^{\log_b a} = 1$$

Master Theorem Case 3 applies.
Therefore:

$$\boxed{T(n) = \Theta(n)}$$

4. Space Complexity
The recursion depth is:

$$\log_2 n$$

Each recursive call stores only constant information.
Hence:

$$\boxed{Space\ Complexity = O(\log n)}$$

Final Answer
Time Complexity:

$$\boxed{\Theta(n)}$$

Space Complexity:

$$\boxed{O(\log n)}$$

Conclusion: The cloud backup analysis algorithm performs linear total work because the recursive processing forms a geometric series, while the recursive stack requires logarithmic space.